

AI-Powered Real-Time Log Monitoring and Incident Response System for Web Applications

Project description

- System detects anomalies in real-time logs for early failure identification.
- Dashboard displays alerts and insights for faster incident response.

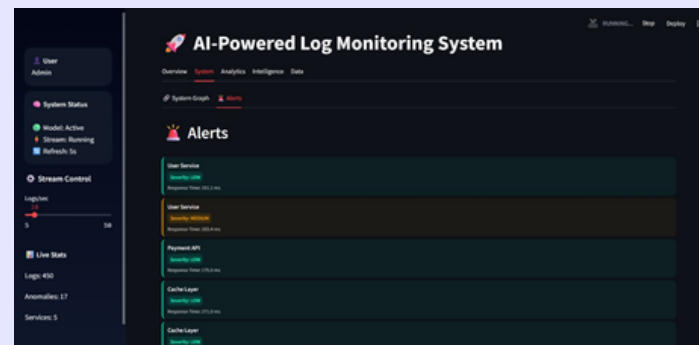
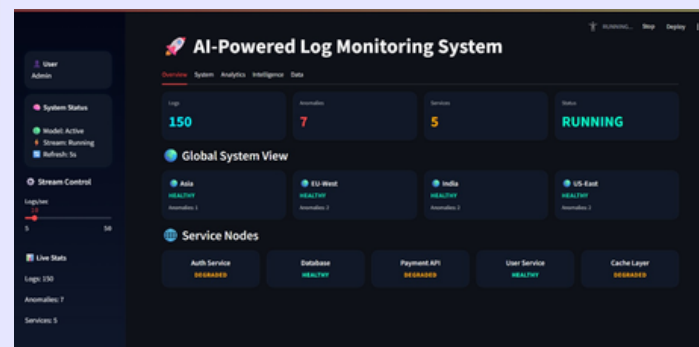
Objective

- Machine learning is used to detect anomalies in log data for early identification of system issues.
- System aims to improve incident response and enhance reliability through real-time monitoring.

Methodology

- Logs are collected from web servers, APIs, and backend services and preprocessed to remove noise and standardize formats.
- Key features such as timestamps, error rates, and service identifiers are extracted.
- Machine learning models analyze these patterns to detect anomalies in real time. Detected anomalies are correlated across components to identify related incidents, and alerts are generated and displayed through a monitoring dashboard for timely response.

RESULTS



Challenges

- Handling high volumes of structured log data in real time is challenging and requires efficient processing.
- Ensuring accurate anomaly detection with low false positive correlating events across distributed systems remains difficult.

Conclusion

The proposed system demonstrates an effective approach to real-time log monitoring using machine learning for anomaly detection and incident identification. By analyzing continuous log streams, it enables early detection of system issues and improves response time. The use of event correlation helps in understanding related incidents across components. Overall, the system enhances reliability, reduces operational effort, and provides a scalable solution for modern web applications.

References

- X. Zhao, Z. Jiang and J. Ma, "A Survey of Deep Anomaly Detection for System Logs," 2022 International Joint Conference on Neural Networks (IJCNN), Padua, Italy, 2022, pp. 1-8, doi: 10.1109/IJCNN55064.2022.9892726. <https://ieeexplore.ieee.org/document/9892726>
- M. Landauer, S. Onder, F. Skopik and M. Wurzenberger, "Deep learning for anomaly detection in log data: A survey," Machine Learning with Applications, vol. 12, p. 100470, 2023, doi:10.1016/j.mlwa.2023.100470. <https://www.sciencedirect.com/science/article/pii/S2666827023000233>